

The Body Beautified

A largish step short of the cyborg, the Bionic Man is already a reality

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Here we go again about man-machine convergence: two years ago, we spoke about cyborgs and the engineering of the human condition. This time it's about bionics. So why are we repeating ourselves?

We aren't. "Cyborgism" is about transcending human capabilities, about creating supermen. Bionics is about repairing injured or disabled humans so they can function normally—as normally as possible, at any rate. Becoming a bionic man may not sound as cool as becoming a cyborg, but the stories are just as fascinating.

"Bionics" actually refers to the application of mechanisms found in nature to the design of artificial systems using technology. However, there is so much usage of the term in the context of designing body parts that "bionics" is now almost synonymous with "artificial organs".

Humans are benefiting from bionics to a degree of sophistication you wouldn't believe. Though expensive, there are bionic eyes, ears, arms, legs, knees, lungs, livers, you name it... a bionic anus has been developed, too, in case you were wondering. The following, then, is the story of how technology is repairing bodies and restoring lost bodily functions—a story of brain, mind, body, machine.

Sight And Sound

Research into the possibilities of bionic enhancements dates back a long time, but the first real breakthrough—at least, the most-remembered and best-known achievement of the early days—was the development of a bionic ear. The Australians are so (rightly) proud of it, it has been featured on a postage stamp: they call it the "Australian Bionic Ear."

Developed by Prof Graeme Clark, the prototype was first implanted in an adult at The Royal Victorian Eye and Ear Hospital in 1978. This one let the patient hear speech for the most part: speech, aided by a small processor that the patient could wear, was analysed and converted to electrical signals that were direct stimulation to the main hearing nerve.

The bionic ear has been brought much closer to perfection over the years. It is now more popularly known as the cochlear implant.

In the case of the sense organs, pretty much the same principle applies throughout all implants: how do we see, and how do we hear? Information is converted into electrical impulses that speed along nerves. The same thing happens in the Argus II system, which consists of external glasses, a retinal implant consisting of electrodes, and a wearable processor. The camera on the glasses views the image and sends signals to the processor. These are converted to bio-friendly information, which is sent back to the glasses and then wirelessly to a receiver under the surface of the eye. Once the signals are inside the eye, it's easy to imagine: the signal is conveyed to the electrodes in the retinal implant, which stimulate the retina appropriately. (The retina then does its job of sending signals to the optic nerve, which carries on to the brain.)

Note that both systems mimic biology: they have been inspired by the appropriate biological mechanisms, and they do almost exactly the same thing their biological counterparts do.

Argus II has been given the go-ahead by the US Food and Drug Administration, and may be used in patient trials—initially with patients over the age of 50. Commercialisation will happen if successful, and the system will cost around \$30,000 (Rs 13.5 lakh). It's obviously not for everybody—yet.

It's fine in theory, but what do patients experience? It's encouraging, but far, very far, from perfect. Terry Byland, then 58, was fitted with an implant in 2004. When it was done, "It was like seeing assembled dots—now it's much more than that," Byland says. "When I am walking along the street I can avoid low-hanging branches—I can see the edges of the branches." What about faces? "I can't recognise faces, but I can see them like a dark shadow." What this means is that we're still a long way from "restoring vision"—it's more like helping the blind live their lives without someone (or a dog) having to guide them all the time.

An Arm And A Leg—And A Mind

Jesse Sullivan can put on his socks, eat without assistance, and vacuum the floor without assis-

